



KENYA MEDICAL RESEARCH INSTITUTE

State Corporation under the Ministry of Health | Est. 1979
In Search of Better Health

PATHWAY TECHNOLOGIES | PROOF OF CONCEPT — COMPLETE SYSTEM DESIGN
DOCUMENT

KEMRI INTEGRATED MONITORING & EVALUATION SYSTEM KIMES

COMPLETE WORKFLOW · GOVERNANCE · POST-STUDY OUTPUT CAPTURE
DONOR LIAISON & CONCURRENT REPORTING · ERP INTEGRATION · NON-CONFORMITY
ESCALATION

WITH SAMPLE DASHBOARDS & DATA VISUALISATIONS FOR ALL STAKEHOLDERS

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Complete PoC — v5.0 Merged + Visualisations	Version 5.0 May 2026	Pathway Technologies for KEMRI	PoC Draft — For Review & Comment

This Document — KIMES v5.0 with Visualisations

This is the single, authoritative KIMES design document for KEMRI, incorporating: (a) the complete 15-step end-to-end workflow; (b) governance and peer-led review structures; (c) donor liaison and concurrent reporting framework; (d) ERP Procurement and HR integration nodal models; (e) non-conformity escalation protocol including donor notification; and (f) sample dashboards and data visualisations for PI, Centre Director, and Board audiences — all presented with the official KEMRI logo.



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1. INTRODUCTION & CONTEXT

1.1 Background — The Blind Spot

Kenya Medical Research Institute (KEMRI) currently uses a paper-based Research Implementation and Grant Monitoring Form (Version 05, March 2026) as its primary instrument for tracking research projects. A structural gap means Principal Investigators (PIs) largely report to external donors, but not systematically to KEMRI management and Board. As a result:

- The Board has no real-time visibility of what research is being conducted, its status, or its outputs
- Centre Directors and senior management cannot aggregate performance data across projects
- Reports submitted to donors are rarely shared internally, preventing institutional learning
- Post-study outputs — publications, conference presentations, policy briefs, patents — are not systematically captured
- Research uptake and policy impact is largely unmeasured
- Donor reporting and institutional reporting operate in parallel silos — creating information asymmetry and accountability gaps

1.2 KIMES Purpose

The KEMRI Integrated Monitoring and Evaluation System (KIMES) is designed to close all of these gaps through a structured digital reporting workflow that:

- (i) requires PIs to report milestones against agreed objectives at defined intervals;
- (ii) routes reports through a scientifically credible peer-led review structure;
- (iii) surfaces consolidated performance data to management and the Board in real time;
- (iv) captures the full lifecycle of research outputs;
- (v) ensures the donor, the PI, and KEMRI management all receive the same verified information simultaneously; and
- (vi) integrates with KEMRI's ERP system to provide real-time procurement and HR visibility per project.

1.3 Review Authority — The Professional Hierarchy

Critical Design Decision: Peer-Led Review, Not Junior-Led Gatekeeping

PIs are reviewed by Centre Directors or Senior Scientists at equal or higher professional rank — never by junior M&E Officers. M&E Officers perform two functions only: (1) DATA QUALITY FACILITATION — the automated DQA engine and M&E Officers ensure submitted data is complete and correctly entered; they do not judge scientific quality. (2) SYSTEM FACILITATION — M&E Officers administer KIMES, send reminders, prepare data packages for Centre Director review, and manage the AI reporting pipeline. All scientific review, RAG assignment, and escalation authority rests with Centre Directors, the DG, and the Board.



2. GOVERNANCE & REVIEW AUTHORITY STRUCTURE

2.1 Review Authority Matrix

Workflow Stage	Responsible Party	Authority Level	M&E Officer Role
Project registration & logframe approval	Centre Director / Scientific Committee	Scientific & institutional	Facilitate form, check completeness, assign project ID
KPI & milestone plan validation	Centre Director + MEL Division Head	Joint: scientific + systems	Ensure indicators are SMART and system-compatible; no veto on scientific design
Quarterly milestone data entry	Principal Investigator	PI solely responsible	Technical support, pre-fill known fields, send reminders
Automated DQA	KIMES System (automated)	Technical / system-level	Monitor DQA scores; follow up on flagged fields; no scientific judgment
Scientific peer review & RAG assignment	Centre Director or Senior Scientist (equal/higher rank)	Scientific peer authority	Prepare data summary package; no independent review or RAG authority
Escalation for non-reporting PIs	MEL Head → Centre Director → DG → Board	Progressive institutional authority	Initiate escalation log; draft notices; route through correct channels
Donor concurrent report routing	Grants Management Office + MEL Division	Institutional + grant management	Configure donor portal; manage report calendar; route approved reports
Procurement service requests	PI (initiator) + ERP Procurement (processor)	PI initiates; Procurement processes	KIMES-ERP API management; procurement status monitoring
HR service requests	PI (initiator) + Centre Director (endorser) + ERP HR (processor)	Centre Director endorses; HR processes	KIMES-ERP API management; staffing dashboard
Post-study output capture	PI (primary) + M&E Officer (facilitation)	PI responsibility; system auto-imports where possible	Maintain output registry; DOI auto-import; citation auto-tracking
Donor non-conformity notification	Director General (exclusively)	Executive — DG only	Compile KIMES audit trail; draft DG letter from template; route for DG approval

2.2 What M&E Officers DO and DO NOT Do

What M&E Officers DO

- System administration and user support for KIMES across all centres
- Send automated reminders to PIs at 30, 14, and 7 days before reporting deadlines



- Monitor the automated DQA engine and follow up with PIs on data entry errors — without judging scientific content
- Prepare consolidated data packages for Centre Directors to use in their peer review
- Manage the escalation log and route non-compliance notices through correct channels
- Facilitate AI report generation and prepare drafts for Centre Director or DG approval
- Operate the GIS platform; manage KIMES-ERP integration monitoring; manage Donor Portal

What M&E Officers DO NOT Do

- Approve or reject any PI's scientific progress report
- Assign or change RAG status independently
- Query the scientific content or methodology of a PI's research
- Initiate or send any escalation notice to a donor — that is a DG-only function

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3. END-TO-END WORKFLOW — ACTIVE STUDY PHASE (Steps 1–9)

The following nine workflow steps cover the active project lifecycle from registration through to post-study transition. Steps 10–15 in Section 4 cover post-study output capture and long-term tracking.

1

Pre-Study

Project Registration & Logframe Setup | *PI (primary) · Centre Director (approver)*

PI registers the project in KIMES upon grant award notification. The system generates a unique Project ID and a logframe shell pre-populated with the grant account number, funder details, study type, contract number, SERU approval number, and proposed start/end dates. PI completes project-specific objectives (up to 5), implementation sites with GPS coordinates, and co-investigator details.

Inputs	Outputs	Trigger / SLA	Escalation
▶ Signed grant agreement ▶ SERU/NACOSTI approval letter ▶ KEMRI Form Sections 1–3 data	▶ Unique KIMES Project ID ▶ Logframe shell (editable) ▶ GIS site pin created ▶ Project visible on Management Dashboard ▶ Donor Portal record created	▶ New grant award notification ▶ PI self-initiates within 14 days of award	See Sec. 7

2

Pre-Study

KPI & Milestone Plan Definition | *PI (primary) · Centre Director (approver) · MEL Division (support)*

PI defines project-specific KPIs, expected outputs per quarter, milestone dates, financial budget breakdown, and data collection methods. Centre Director or Senior Scientist reviews and approves the monitoring plan. M&E Officers provide technical support to ensure indicators are SMART and system-compatible — no veto authority.

Inputs	Outputs	Trigger / SLA	Escalation
▶ Approved logframe ▶ Funder monitoring plan (if available) ▶ KIMES indicator registry templates	▶ Approved project monitoring plan ▶ KPI targets locked in KIMES (version-controlled) ▶ Baseline recorded ▶ Reporting schedule auto-generated ▶ Donor Portal reporting calendar configured	▶ Step 1 completion ▶ Centre Director logframe approval	See Sec. 7

3

Active Study

Quarterly Milestone Reporting by PI | *PI solely responsible*

At 30, 14, and 7 days before each quarter-end deadline, KIMES sends automated reminders to the PI. The PI logs quarterly milestone data: KPI achievement vs targets, financial utilization, staff status, lab analyses summary, equipment acquired, capacity



building activities, and emerging risks. The AI engine pre-fills fields already known to KIMES. Submission simultaneously CC's to Centre Director, DRD, Grantsmanship, and PMME — closing the loop where PIs previously only reported to funders.

Inputs	Outputs	Trigger / SLA	Escalation
▶ AI pre-filled KIMES data ▶ Previous quarter report ▶ KPI targets from Step 2 ▶ Financial data from ERP/IFMIS	▶ Quarterly milestone report submitted ▶ Timestamped read-only record ▶ Auto-CC to Centre Director, DRD, Grantsmanship, PMME ▶ DQA score calculated ▶ Donor Portal report queue updated	▶ Quarter-end automated reminders: D-30, D-14, D-7	See Sec. 7

4

Active Study

Automated Data Quality Assessment (DQA) | KIMES System (automated) · M&E Officer (follow-up support)

Immediately upon PI submission, the KIMES automated DQA engine runs eight validation checks: required field completeness ($\geq 85\%$), numeric range plausibility, GPS coordinate validation, financial arithmetic consistency, date logic, cross-field consistency, SERU expiry date check, and duplicate submission detection. Reports below 85% completeness are automatically returned to the PI with specific field-level flags.

Inputs	Outputs	Trigger / SLA	Escalation
▶ PI submitted quarterly report ▶ KIMES validation rules ▶ Project monitoring plan (for range checking)	▶ DQA score (0–100%) per report ▶ Field-level validation flags ▶ Auto-return to PI if score $< 85\%$ ▶ DQA log entry in KIMES audit trail ▶ SERU expiry alert if within 60 days	▶ Automatic: triggered at PI submission	See Sec. 7

5

Active Study

Centre Director Scientific Peer Review & RAG Assignment | Centre Director (or designated Senior Scientist at $\geq$ PI rank)

The Centre Director (or a Senior Scientist at equal or higher rank — never a junior M&E Officer) reviews the DQA-validated report. The Centre Director may:

- Accept,
- Query — send a specific question to the PI, or
- Escalate.

The Centre Director assigns RAG status: Green (on track), Amber (minor variance, mitigation acceptable), Red (significant deviation, remedial action required).

Inputs	Outputs	Trigger / SLA	Escalation
▶ DQA-validated PI report	▶ Approved, queried, or escalated report	▶ DQA score $\geq 85\%$ and PI corrections accepted	See Sec. 7



- ▶ Previous quarter RAG status
- ▶ Approved monitoring plan
- ▶ MEL Officer data summary package

- ▶ RAG status assigned (Green/Amber/Red)
- ▶ Centre Director comments visible to PI
- ▶ Management dashboard updated
- ▶ Donor Portal updated with approved report

6

Active Study

Escalation Protocol for Non-Reporting PIs | MEL Head (initiator) → Centre Director → DG → Board

Day 1 post-deadline: automated system reminder. Day 7: MEL Division Head sends written notice to PI, CC to Centre Director. Day 14: Centre Director formally notifies PI; DG Office copied. Day 21: Non-reporting projects flagged on Board Dashboard. See Section 7 for full non-conformity classification and donor notification protocol.

Inputs	Outputs	Trigger / SLA	Escalation
▶ Missed reporting deadline (system-detected) ▶ Prior escalation history ▶ PI contact details from HR system	▶ Escalation log entry (timestamped, immutable) ▶ Formal notices at D+7, D+14 ▶ Board Dashboard flag at D+21 ▶ Full audit trail for HR/performance use	▶ PI failure to submit within 15 days of quarter-end	See Sec. 7

7

Active Study

AI Narrative Report Generation | KIMES AI Engine (draft) · MEL Division (prep) · Centre Director (approval)

Following Centre Director approval, the KIMES AI Reporting Engine (RAG-constrained — reads only from KIMES, never fabricates) auto-drafts the quarterly progress narrative. Produces Word (.docx), Excel (.xlsx), and PDF simultaneously. Donor receives their concurrent version in their required template format. No AI-generated report is distributed without Centre Director or DG sign-off.

Inputs	Outputs	Trigger / SLA	Escalation
▶ Centre Director-approved PI report ▶ KIMES indicator data ▶ KIMES financial data (from ERP) ▶ Funder report template	▶ AI-drafted report in Word (.docx) ▶ Formatted data export in Excel (.xlsx) ▶ Final signed report in PDF ▶ Donor-formatted report sent to Donor Portal ▶ Version-controlled report in KIMES document store	▶ Centre Director approval of PI report (Step 5)	See Sec. 7



8

Active Study

Donor Concurrent Report Routing | *PI (primary) · Grants Management Office · MEL Division (routing)*

When a PI submits any report to a donor, KIMES captures this through a simple check-in step. The system automatically timestamps the submission, routes the KIMES-generated concurrent version to the Donor Portal, and sends notification to Centre Director, DRD, Grantsmanship, and PMME. PIs retain full ownership of the funder relationship.

Inputs	Outputs	Trigger / SLA	Escalation
- PI-prepared donor report - Funder submission deadline from grant agreement - Centre Director-approved KIMES report	- KIMES submission timestamp (audit trail) - Donor Portal updated with approved report - Notification to: Centre Director, DRD, Grantsmanship, PMME - Compliance record updated	PI funder reporting deadline (per grant agreement)	See Sec. 7

9

Active Study

GIS & Spatial Layer Update | *GIS Specialist (primary) · M&E Officer (data sync support)*

Field GPS data from ODK Collect and KoboToolbox syncs automatically to the KIMES GIS layer. The GIS Specialist updates project site maps, disease burden overlays, and county-level research coverage maps. Maps are embedded in quarterly reports, the Management Dashboard, and the Board Scorecard.

Inputs	Outputs	Trigger / SLA	Escalation
- GPS data from ODK/KoboToolbox - Declared study site polygons - Disease burden data from DHIS2 - Satellite imagery (Sentinel-2)	- Updated project site maps (all 47 counties) - Disease burden overlay maps - Research coverage map per county - Spatial analysis reports (for relevant studies)	- PI quarterly report submission (Step 3) - Field data collection sync (continuous)	See Sec. 7



4. POST-STUDY OUTPUT CAPTURE & LONG-TERM TRACKING (Steps 10–15)

Design Principle — Long-Term Output Capture

Research impact often materializes 1–5 years after a project closes. KIMES is designed to track this full trajectory. PI-linked output records remain active for 7 years post-project closure. The AI engine cross-references DOIs, PubMed IDs, and policy document citations annually to auto-update impact metrics without requiring PI data re-entry.

10

Post-Study

Study Closure Report & Transition to Post-Study Tracking | *PI (primary) · Centre Director (approver)*

At project end, PI submits a structured Study Closure Report covering: final KPI achievement, final financial reconciliation, sustainability plans, data sharing status, sample storage details, and a preliminary list of expected outputs. KIMES automatically transitions the project to 'Post-Study Tracking' status — the PI's output obligations continue for 7 years.

Inputs	Outputs	Trigger / SLA	Escalation
▶ Final quarterly report ▶ Final financial report ▶ Funder final report ▶ SERU closure notification	▶ Approved Study Closure Report ▶ Project status: Active → Post-Study Tracking ▶ Post-Study Output Registry created for PI ▶ Data sharing compliance record	▶ Project end date reached	<i>See Sec. 7</i>

11

Post-Study

Research Abstracts & Conference Presentations | *PI (self-reporting) · M&E Officer (registry management)*

PI logs via the 'My Research Outputs' portal: abstract title, conference/event name, type (oral/poster/keynote/webinar/symposium), date, location, co-authors, acceptance status, and links to abstract repository. For major international conferences, the Communications team receives an auto-notification.

Inputs	Outputs	Trigger / SLA	Escalation
▶ Abstract submission or acceptance confirmation ▶ Conference programme listing (optional upload)	▶ Abstract/presentation record in Post-Study Output Registry ▶ Comms team notified for major international conferences ▶ Annual conference impact tally auto-updated	▶ PI-initiated: at submission, acceptance, or presentation	<i>See Sec. 7</i>

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Publications — Submission, Acceptance & Citation Tracking | *PI (self-reporting) · KIMES System (auto-import via DOI/PubMed API)*



Post-Study

KIMES captures publications at three stages: (1) On Submission — PI logs manuscript title, journal, submission date. (2) On Acceptance/Publication — PI enters DOI; system auto-imports full citation metadata from Crossref API. (3) Ongoing — KIMES connects to Scopus/PubMed APIs to auto-update citation counts quarterly. High-impact publications (IF>5) trigger an automated notification to Communications and DG Office.

Inputs	Outputs	Trigger / SLA	Escalation
- Manuscript submission confirmation - DOI (on publication) - Crossref/PubMed API (auto) - Scopus/Web of Science API (auto)	- Publication record in Post-Study Output Registry - Full citation metadata (auto-imported via DOI) - Quarterly citation count updates (auto) - High-impact alert to Comms & DG (IF>5)	- PI-initiated on submission - Auto-triggered on DOI detection	See Sec. 7

13

Post-Study

Datasets, Tools & Research Materials — Open Science Registry | PI

(self-reporting) · Data Management Officer (verification)

PI logs all data deposits: dataset name, repository, DOI, access level (Open/Restricted/Controlled), embargo period, and date of deposit. FAIR data compliance score (Findable, Accessible, Interoperable, Reusable) calculated per dataset.

Inputs	Outputs	Trigger / SLA	Escalation
- Completed research datasets - Repository access credentials - Project data sharing plan - Funder open access requirements	- Dataset deposit record in KIMES registry - FAIR compliance score per dataset - Annual data deposit report for Board	- PI-initiated at time of data deposit 12 months post-closure: auto-reminder	See Sec. 7

14

Post-Study

Policy Briefs, Guidelines & Evidence Uptake Tracking | PI (self-reporting) · MEL Division (annual AI scan)

PI and Centre Director log: any policy brief produced, any clinical guideline citing KEMRI research, any technical assistance provided to government, any media engagement. MEL Division runs an annual AI policy citation scan against publicly available policy documents, WHO reports, and national health strategies. Annual Research Uptake Score (1–10) produced per programme.

Inputs	Outputs	Trigger / SLA	Escalation
- Policy briefs produced - Guideline documents citing KEMRI research - Annual AI policy citation scan results	- Policy engagement log entry - Annual Research Uptake Score per programme - Policy uptake section in Board Annual Report - AI-generated policy impact narrative	- PI-initiated on any policy engagement event - Annual AI scan: each August	See Sec. 7



15

Post-Study

Intellectual Property, Patents & Technology Transfer | *PI + Innovation Unit (joint) · Legal/IP Office (verification)*

PI and Innovation Unit jointly log: IP asset type, IP title, filing date, patent number, jurisdiction, commercialization stage (using KEMRI's 10-stage framework), licensing arrangements, and revenue generated. The system auto-monitors patent expiry dates.

Inputs	Outputs	Trigger / SLA	Escalation
▶ Patent filing confirmation ▶ IP registration certificates ▶ Licensing agreements ▶ Commercialization stage assessment	▶ IP asset record in KIMES registry ▶ Commercialization stage tracked (10-stage) ▶ Patent expiry auto-alert (12 months before) ▶ Annual IP portfolio report for Board	▶ PI + Innovation Unit: on filing or registration ▶ Annual: IP portfolio audit	See Sec. 7



5. DONOR LIAISON & CONCURRENT REPORTING FRAMEWORK

5.1 Concurrent Reporting Architecture — Four Tiers

Defining Concurrent Reporting

Concurrent reporting does not mean three separate reports are produced. It means a single verified KIMES data record generates multiple formatted outputs simultaneously — one for the donor (in their required format), one for KEMRI management (institutional dashboard), and one for the Board (scorecard). The PI validates the data once. KIMES produces all outputs. All parties see the same verified facts at the same moment.

Tier	Mechanism	What Donor Receives	What KEMRI Receives	Timeline
Tier 1 Real-Time Portal	Donor issued read-only KIMES credentials at project inception: live project status, KPI gauges, SERU compliance, budget utilization	Live dashboard data (read-only). Same data Centre Director sees.	Same data the donor sees. No information asymmetry.	Continuous from project registration
Tier 2 Concurrent Quarterly	PI submits → Centre Director approves → KIMES simultaneously generates KEMRI institutional report AND funder-formatted report	AI-drafted report in donor's template format — Word and PDF — within 24 hours of approval	Institutional quarterly report on management dashboard — same data	Within 24 hours of Centre Director approval
Tier 3 Annual Financial	KIMES extracts reconciled financial statement annually from IFMIS/ERP integration, shared simultaneously with donor and KEMRI Finance	Annual financial statement: budget vs actual vs balance, in required currency	Same financial data in institutional format. No separate preparation by Finance.	By 31 July each year (end of GOK FY)
Tier 4 Final Report	On project closure, KIMES generates final report simultaneously for donor submission, KEMRI institutional record, and Board annual report contribution	Final project report in donor template — all sections from KIMES closure data	Institutional project closure record. Transition to Post-Study Output Registry.	Within 30 days of project end date

5.2 Donor Engagement Protocol

5.2.1 Grant Onboarding Protocol

At the point of grant award, the Grants Management Office initiates a Grant Onboarding meeting with the donor covering:

- Confirmation of reporting calendar — entered into KIMES as the automated reminder schedule
- Identification of the donor's reporting template and format requirements — uploaded to KIMES as the AI report template
- Agreement on the donor's KIMES portal access credentials and data sharing scope



- Agreement on concurrent reporting as the institutional standard
- Identification of the donor's designated contact officer for escalation purposes (Section 7)
- Confirmation of financial reporting currency, audit requirements, and incremental funding protocols

5.2.2 Grant Agreement Data Addendum

KEMRI will attach a standard Grant Agreement Data Addendum to all new grant agreements establishing: KEMRI's right to share project performance data with its Board; the concurrent reporting obligation; the escalation protocol including circumstances for DG donor notification; and data sovereignty provisions.

Reporting Event	PI Deadline in KIMES	Centre Director Review	AI Report Generation	Donor Submission	KEMRI Record Update
Quarterly Progress Report	15 days after quarter-end	D+5 after PI submission	D+6 (within 30 min of approval)	D+7 — submitted by Grants Office	D+7 — dashboard updates simultaneously
Semi-Annual Report	30 days before donor deadline	5 days after PI submission	Within 30 min of approval	On donor deadline — never before approval	Same day as donor submission
Annual Progress Report	45 days before donor deadline	7 days after PI submission	Within 1 hour of approval	On donor deadline	Same day as donor submission
Final Report	60 days before project end	10 days after PI submission	Within 2 hours of approval	On donor deadline	Triggers Step 10 project closure workflow



6. ERP INTEGRATION — PROCUREMENT & HUMAN RESOURCES

Integration Architecture Principle

KIMES is not designed to replace the KEMRI ERP system. The ERP (IFMIS or equivalent) remains the system of record for all financial transactions, procurement contracts, and HR records. KIMES is the Research Intelligence Layer that sits above the ERP — contextualizing ERP data within each research project, and feeding project-specific service requests back to the ERP as structured API calls.

6.1 Procurement Integration — 7-Node Pipeline

Node	Actor	Action	Data TO KIMES	Data TO ERP
Node 1 Procurement Plan	PI + MEL Officer	PI creates Procurement Plan in KIMES: item name, specification, quantity, estimated cost, required delivery date, budget line, and justification.	Procurement Plan record in KIMES project file. Visible on management dashboard under Planned Procurement.	Procurement Plan exported as structured JSON/XML requisition to ERP Procurement module — pre-populating the ERP purchase requisition with project number, budget line, and specifications.
Node 2 Purchase Requisition	PI (initiator) + Procurement Officer	ERP Procurement Officer receives KIMES-generated requisition. Validates budget availability and creates a formal Purchase Requisition (PR) in the ERP.	ERP pushes PR number back to KIMES, linked to originating project. KIMES updates: Procurement Plan Item — PR raised [PR-YYYY-NNN].	Procurement Officer processes PR in ERP following standard regulations (PPADA 2015 thresholds).
Node 3 Approval & Tendering	Procurement Committee	PR approved through ERP workflow. Proceeds to supplier selection via LPO, RFQ, or open tender.	ERP pushes approval status and tendering method to KIMES. Project procurement item updated: Approved — Tendering method — Expected award date.	All tendering and supplier selection conducted in ERP per PPADA 2015.
Node 4 LPO / Contract Award	Head of Procurement	ERP generates LPO or contract. Supplier notified. Delivery timeline confirmed.	ERP pushes LPO number, supplier name, contracted amount, and expected delivery date to KIMES. Budget commitment recorded in KIMES financial dashboard.	All supplier payment terms and contract management in ERP.



Node	Actor	Action	Data TO KIMES	Data TO ERP
Node 5 Delivery & GRN	PI (receiver) + Procurement Officer	PI confirms receipt in KIMES: item, quantity, condition, tag number, delivery date. Procurement Officer confirms GRN in ERP.	PI delivery confirmation logged in KIMES linked to project. Equipment tag number and location recorded (digitizing Form Q39). KIMES updates to: Received — GRN number.	Procurement Officer logs GRN in ERP, releasing payment to supplier.
Node 6 Payment & Close-Out	Finance Officer	ERP Finance processes payment to supplier against GRN. Payment recorded against project budget line.	ERP Finance pushes payment confirmation and actual amount paid to KIMES. Budget utilization rate recalculated. Procurement item marked Closed.	Payment processed in ERP as system of record.
Node 7 Asset Registry	MEL Officer / PI	For capital items, KIMES automatically creates an Equipment Registry entry from delivery confirmation data.	Equipment Registry updated in KIMES. Visible on project record and institutional asset dashboard.	Asset value posted to ERP Fixed Assets register via API.

6.2 Human Resources Integration — 7-Node Pipeline

Node	Actor	Action	Data TO KIMES	Data TO ERP/HR
Node HR-1 Staffing Plan	PI	PI creates Project Staffing Plan at registration: roles required, number of staff, required qualifications, funding source, start/end dates, and role codes (Form Q36/Q37 taxonomy — now digital).	Staffing Plan record in KIMES project file. Linked to project budget (personnel line). Visible to Centre Director and HR Division.	Staffing Plan exported to HR module as structured staffing request — providing HR with full project context.
Node HR-2 Recruitment Request	PI (initiator) + HR Division	PI initiates recruitment or secondment request in KIMES for each position. KIMES generates structured requisition to HR.	HR request logged in KIMES with unique Request ID. Status visible: Draft / Submitted / Under HR Review / Approved / Filled.	HR requisition transmitted to ERP HR module with project number, role code, qualification requirements, duration, and project funding confirmation.
Node HR-3 Contract Issuance	HR Division	HR processes request, advertises post (if new hire),	ERP HR pushes contract issuance notification to KIMES: employee name,	Contract issued in ERP as system of



Node	Actor	Action	Data TO KIMES	Data TO ERP/HR
		shortlists, interviews, selects, and issues contract.	staff ID, role, start date, contract end date, payroll number, and grading. KIMES project staffing plan item updated to Filled.	record. Payroll set up in ERP.
Node HR-4 Active Monitoring	ERP (automated daily sync)	ERP HR syncs daily with KIMES: current staff active on each project, contract end dates, leave records, and payroll charges to project budget lines.	KIMES project record shows: active staff count by role code, staff names, contract end dates, leave taken, and payroll charges to project.	Staff management (leave approval, performance, disciplinary) remains entirely in ERP HR — KIMES reads only.
Node HR-5 Contract Extension	PI (initiator) + Centre Director (endorser) + HR	PI initiates extension request in KIMES. Centre Director endorses within KIMES. KIMES transmits endorsed request to ERP HR.	Extension request logged in KIMES with Centre Director endorsement timestamp. Status updated when HR processes.	HR processes extension in ERP. New contract end date pushed back to KIMES.
Node HR-6 Capacity Building	HR Division + PI	All formal training, professional development, and academic qualifications recorded in ERP HR are synced to KIMES.	Training records synced to KIMES: course name, type, dates, cost, institution, and outcome. Appears in project Q38 data and institutional capacity dashboard.	Training approvals and costs remain in ERP HR.
Node HR-7 Staff Departure	HR Division	When a staff member's contract ends or they resign, ERP HR triggers a departure notification to KIMES.	Project staffing record updated: role marked Vacant. PI and Centre Director alerted. KIMES flags project with potential capacity risk if a key role is vacant.	Off-boarding processed in ERP as standard.



7. NON-CONFORMITY ESCALATION PROTOCOL — INCLUDING DONOR NOTIFICATION

Foundational Principle — Donor Notification as an Institutional Act

Donor notification of PI non-conformity is always an institutional act — initiated by the Director General on the advice of the Deputy Director (Research) and the Head of MEL, after a formal internal review. It is never initiated unilaterally by an M&E Officer, Centre Director, or any individual staff member. The purpose is proactive accountability management — informing the donor of the situation, demonstrating that KEMRI has taken all appropriate internal measures, and proposing a remediation pathway.

7.1 Non-Conformity Classification Framework

Level	Classification	Definition	KIMES Detection	Internal Authority
Level 1	Minor Non-Conformity	Single late submission (1–14 days); isolated data quality flag; minor budget variance (<10%) without explanation	KIMES automated DQA flag; reminder engine	M&E Officer — automated system action
Level 2	Moderate Non-Conformity	Repeated late submissions (2+ consecutive quarters); DQA score persistently <80%; unexplained budget variance 10–25%	KIMES escalation log: D+7 trigger; DQA trend report	MEL Division Head — formal written notice
Level 3	Significant Non-Conformity	Three or more consecutive quarters non-reported; unexplained budget variance >25%; SERU approval lapsed; staffing gap in critical role >30 days	KIMES escalation log: D+21 Board flag; financial variance alert; SERU expiry alert	Centre Director formal notice; DG Office notified; Board flagged
Level 4	Severe / Donor-Risk Non-Conformity	Four or more consecutive quarters non-reported; evidence of financial mismanagement; grant at material risk	KIMES compliance record: 4+ missed quarters; financial risk flag; DG intervention log	DG formal intervention; Board resolution; Donor notification triggered
Level 5	Institutional Non-Conformity	Centre-wide pattern: multiple PIs non-conforming; systemic data quality failure; institutional financial breakdown	KIMES centre-level compliance dashboard; DQA trend by centre	Board directive; DG action plan; Portfolio-level donor engagement



7.2 Graduated Escalation Protocol — Four Levels

L1

Days 1–14

Automated System Reminders — Minor Non-Conformity | *KIMES System (automated) · M&E Officer (monitoring)*

The KIMES reminder engine sends three automated reminders (D-30, D-14, D-7) before the reporting deadline. If no submission is received by the deadline, KIMES sends a Day 1 post-deadline automated reminder and flags the project on the M&E Officer's daily monitoring queue.

Inputs into KIMES	Outputs from KIMES	SLA / Timeline	Escalation
▶ Missed reporting deadline (system-detected) ▶ PI contact info from HR sync ▶ Prior quarter report (for context)	▶ Three automated email reminders logged in KIMES audit trail ▶ D+1 flag on M&E Officer queue ▶ Project RAG moves to Pending on dashboard	▶ D-30, D-14, D-7: automated reminders ▶ D+1: auto-flag to M&E Officer queue ▶ Resolution window: D+1 to D+14	See Sec. 7

L2

Days 15–30

Formal Written Notice — Moderate Non-Conformity | *MEL Division Head (initiator) · Centre Director (informed)*

If no submission is received by Day 14, the MEL Division Head issues a Formal Written Notice via KIMES internal messaging (with email copy) addressed to the PI, copied to the Centre Director. The notice states facts and obligations without threatening sanctions. The Centre Director is informed but not yet required to act.

Inputs into KIMES	Outputs from KIMES	SLA / Timeline	Escalation
▶ D+14 missed submission (KIMES log) ▶ PI reporting obligation from monitoring plan ▶ Prior escalation history (if any)	▶ Formal Written Notice logged in KIMES (immutable, timestamped) ▶ Centre Director notification ▶ PI response deadline: Day 21 ▶ KIMES compliance record updated: Level 2	▶ MEL Head notice: D+14–15 ▶ PI response window: D+15 to D+21 ▶ Centre Director follow-up call: D+18 (if no PI response)	See Sec. 7

L3

Days 21–45

Centre Director Formal Intervention — Significant Non-Conformity | *Centre Director (lead) · DG Office (notified) · Board (flagged)*

If no submission is received by Day 21, the Centre Director formally intervenes — meets with the PI (documented in KIMES) and issues a Significant Non-Conformity Notice. The DG Office is formally notified. The project is flagged on the Board Dashboard as Non-Compliant. If the PI submits by Day 30, the non-conformity is resolved at Level 3.

Inputs into KIMES	Outputs from KIMES	SLA / Timeline	Escalation
▶ D+21 missed deadline (KIMES log) ▶ Level 2 notice (KIMES record)	▶ Significant Non-Conformity Notice logged in KIMES	▶ Centre Director meeting: by D+23 ▶ DG notification: D+24 ▶ Final deadline to PI: D+30	See Sec. 7



	▸ Centre Director meeting notes ▸ Grant agreement terms	▸ Meeting record in KIMES (date, attendees, outcome) ▸ DG Office notification ▸ Board Dashboard flag: Non-Compliant ▸ Final submission deadline: Day 30	▸ Board Dashboard update: D+21	
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DG Formal Intervention & Donor Notification — Severe Non-Conformity

| *Director General (lead) · Deputy Director Research (advisor) · MEL Division Head (documentation)*

If no submission is received by Day 30 — and this represents a pattern of three or more consecutive non-reporting quarters (or a single instance of financial mismanagement or ethical breach) — the DG initiates Level 4 procedures: (1) DG Formal Letter to PI, (2) Internal Review Board, and (3) Donor Notification if the Internal Review Board recommends (see Section 7.3).

L4
Days 30+

Inputs into KIMES	Outputs from KIMES	SLA / Timeline	Escalation
▸ D+30 missed final deadline (KIMES log) ▸ Complete KIMES non-conformity audit trail ▸ Grant agreement terms and donor contact details ▸ Internal Review Board decision	▸ DG Formal Letter to PI (KIMES template, logged) ▸ Internal Review Board decision record (KIMES) ▸ Donor Notification Letter DG-NCF-001 (if review recommends) ▸ HR formal performance management referral (if recommended) ▸ Board informed at next meeting	▸ DG Formal Letter: D+30–31 ▸ Internal Review Board: D+32–33 ▸ Donor Notification (if approved): D+35–37 ▸ HR referral: simultaneously with donor notification	See Sec. 7

7.3 Donor Notification — Protocol, Content & Format

7.3.1 Preconditions for Donor Notification

Donor notification is only authorized if ALL of the following preconditions are met:

- The project has reached Level 4 Non-Conformity (Day 30+ without submission and an internal review has been completed)
- OR three consecutive quarters of non-reporting at Level 3, despite formal Centre Director intervention
- OR evidence of financial mismanagement, misappropriation, or material budget variance constituting a breach of the grant agreement
- OR a lapsed SERU approval with research activity continuing — an ethical and regulatory violation
- The Internal Review Board has formally recommended donor notification
- The DG has approved the notification
- KEMRI Legal Counsel has cleared the letter for external communication



7.3.2 Donor Notification Letter — Standard Structure (Template DG-NCF-001)

Donor Notification Letter — Six Required Sections

1. Opening — Institutional Context

Formal DG letterhead. Reference: Grant number, project title, PI name. State that KEMRI is writing in its capacity as the institutional host and grant-managing entity, as per the terms of the grant agreement.

2. Statement of the Non-Conformity

A factual, unemotional statement of the specific non-conformity: number of missed reporting periods, dates, any financial variance, any ethical compliance issue. Reference the KIMES audit trail record by document number for evidentiary purposes.

3. Institutional Measures Taken

Enumerate the escalation steps KEMRI has taken: Level 1 automated reminders (dates), Level 2 formal written notice (date, sent by whom), Level 3 Centre Director formal intervention (date, outcome), Level 4 DG formal letter (date). This demonstrates institutional due diligence.

4. Current Status & Grant Risk Assessment

State the current project status from KIMES: KPI achievement, financial utilisation, remaining project period, and KEMRI's assessment of risk level: whether grant objectives are still achievable and whether any funds may be at risk.

5. Proposed Remediation Pathway

Present KEMRI's proposed remediation plan: (a) appointment of an institutional Co-PI or project monitor; (b) a revised reporting schedule with institutional verification at each step; (c) where appropriate, a request for a grant extension or re-phasing. Invite the donor to a joint meeting within 14 days.

6. Commitment to Resolution

Reaffirm KEMRI's commitment to the scientific objectives and the institutional relationship with the funder. State that KEMRI will provide a full status update within 21 days of this letter. Provide DG direct contact details.

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8. SAMPLE DASHBOARDS & DATA VISUALISATIONS

This section presents sample dashboards and data visualizations produced by the KIMES AI Reporting Engine for each stakeholder audience. All data shown is illustrative (Q3 FY2024/25 scenario data) and demonstrates the type of real-time intelligence KIMES will provide to each role.

8.1 PI Project Dashboard

The PI dashboard provides each Principal Investigator with a personalized view of their active projects, KPI progress, procurement items, HR/staffing status, and outputs. This is the primary interface through which PIs submit quarterly reports, track milestone achievement, and manage their project in KIMES.

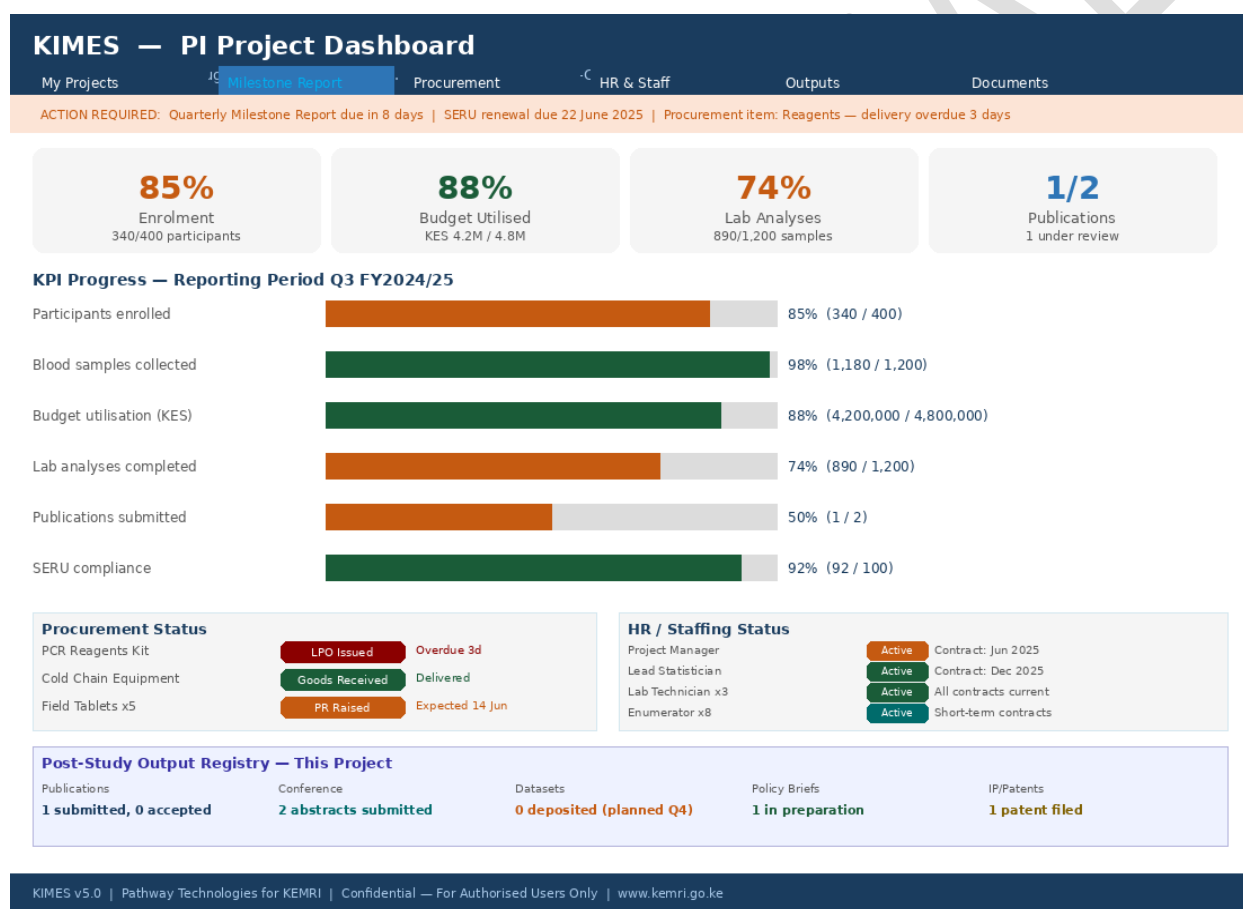


Figure 8.1 — KIMES PI Project Dashboard (Sample: MALM-2024, Dr J. Otieno, CGMR-C, Q3 FY2024/25)

8.2 Centre Director Portfolio Dashboard

The Centre Director dashboard provides a comprehensive portfolio view of all projects within the centre — including RAG status per project, reporting compliance rates, escalation queue requiring Centre Director action, procurement and HR alerts, and cumulative research output tallies. This is the primary interface through which Centre Directors conduct their scientific peer review and RAG assignment.



KIMES — Centre Director Portfolio Dashboard

CGMR-C | Centre for Global Medical Research (Coast) | Kilifi | FY2024/25 Q3

42

Active Projects
35 Green / 5 Amber / 2 Red

83%

Compliance Rate
+25% from Q1 baseline

KES 850M

Total Portfolio
88% utilised

112

Publications YTD
77% of annual target

Project Portfolio — RAG Status Summary

Project ID	PI	Study Type	Budget (KES M)	Q3 KPI%	Compliance	RAG	Action
MALM-2024	Dr Otieno J.	Clinical Trial	48.0	81%	Submitted	Amber	Queried PI
HIVC-2023	Dr Njoroge B.	Cohort Study	32.0	94%	Approved	Green	None
TBDX-2024	Dr Adhiambo S.	Diagnostics	18.5	88%	Approved	Green	None
VACC-2022	Dr Wekesa M.	Vaccine Trial	95.0	76%	Approved	Green	None
EPIO-2024	Dr Karanja F.	Epidemiology	15.0	65%	Approved	Amber	Monitor
DENG-2023	Dr Mwangi P.	Vector Biology	22.0	45%	Overdue 12d	Red	Escalate L2
NUTR-2024	Dr Achieng L.	Nutrition	11.0	82%	Submitted	Green	Review

Escalation Queue (Requires Centre Director Action)

DENG-2023	Dr Mwangi P.	Non-reporting	Day 12 overdue	L2 — MEL Head noticed issued 3 May
SNAP-2022	Dr Kilonzo T.	Budget variance >25%	KES 8.2M unreconciled	L3 — Centre Director to meet PI
MHRT-2023	Dr Ouma S.	SERU lapsed	Expired 28 Apr 2025	L3 — Immediate ethics risk

Centre Research Outputs YTD

89

Publications
(Centre YTD)

45

Conferences
(Centre YTD)

12

Policy Briefs
(Centre YTD)

8

Datasets
(Centre YTD)

3

IP/Patents
(Centre YTD)

6

Policy Citations
(Centre YTD)

Figure 8.2 — KIMES Centre Director Portfolio Dashboard (Sample: CGMR-C, Q3 FY2024/25)



8.3 DG/Board Quarterly Scorecard

The DG/Board Quarterly Scorecard is the most senior dashboard in KIMES. It is auto-generated by the AI Reporting Engine from KIMES data, reviewed by the MEL Division Head, approved by the Director General, and published to the Board Portal within 3 weeks of quarter-end. It provides a comprehensive institutional overview across five sections: KPIs, Strategic Plan progress, Compliance & Governance, Items Requiring Board Attention, and Donor Relationship Summary.

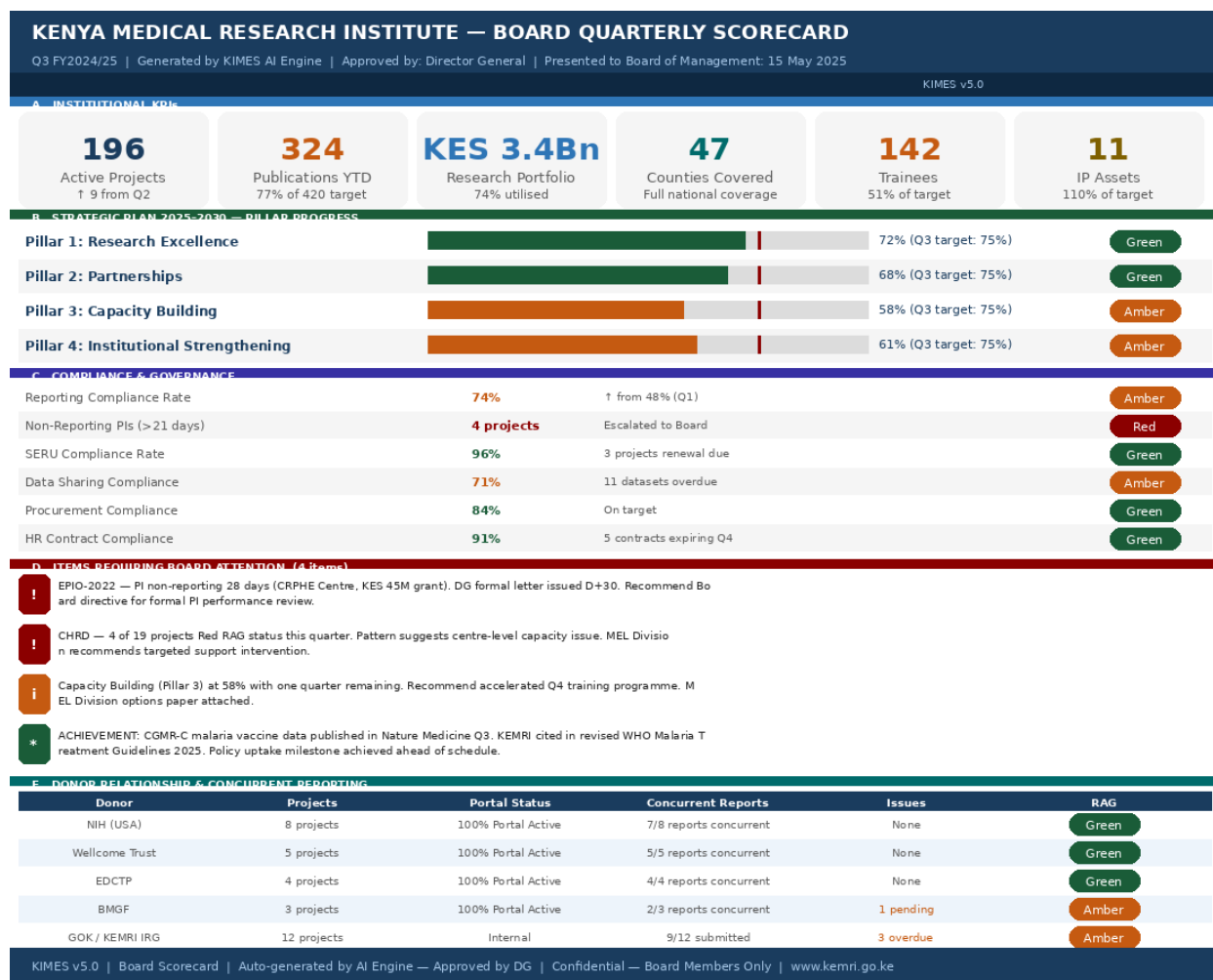


Figure 8.3 — KIMES Board Quarterly Scorecard (Sample: Q3 FY2024/25 — Auto-generated by AI Engine, Approved by DG)



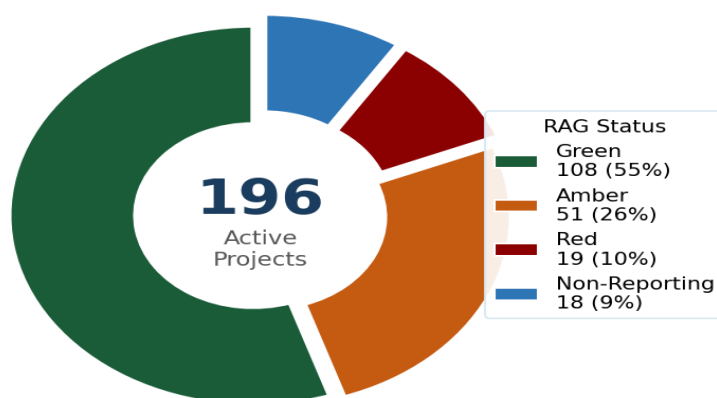
9. DATA VISUALISATIONS — KIMES ANALYTICS

The following charts represent the types of analytical outputs generated automatically by KIMES and embedded in reports for each stakeholder audience. All charts are produced by the KIMES AI Reporting Engine from live KIMES data without any manual data preparation.

9.1 Portfolio RAG Status Overview

The RAG status donut chart is the single most important visual for management and Board audiences — providing an instant read of how the overall research portfolio is performing. It appears on all management dashboards, in all quarterly reports, and as the opening visual of every Board Scorecard.

Portfolio RAG Status — Q3 FY2024/25



Source: KIMES Project Registry | All Centres

Figure 9.1 — Portfolio RAG Status: Active Projects Q3 FY2024/25 (Source: KIMES Project Registry — All Centres)



9.2 Strategic Plan Pillar Progress

This horizontal bar chart maps progress against all four Strategic Plan pillars against the quarterly expected target (75% of annual target by Q3). It is embedded in the Board Scorecard, the DG Executive Dashboard, and the institutional Annual Report.

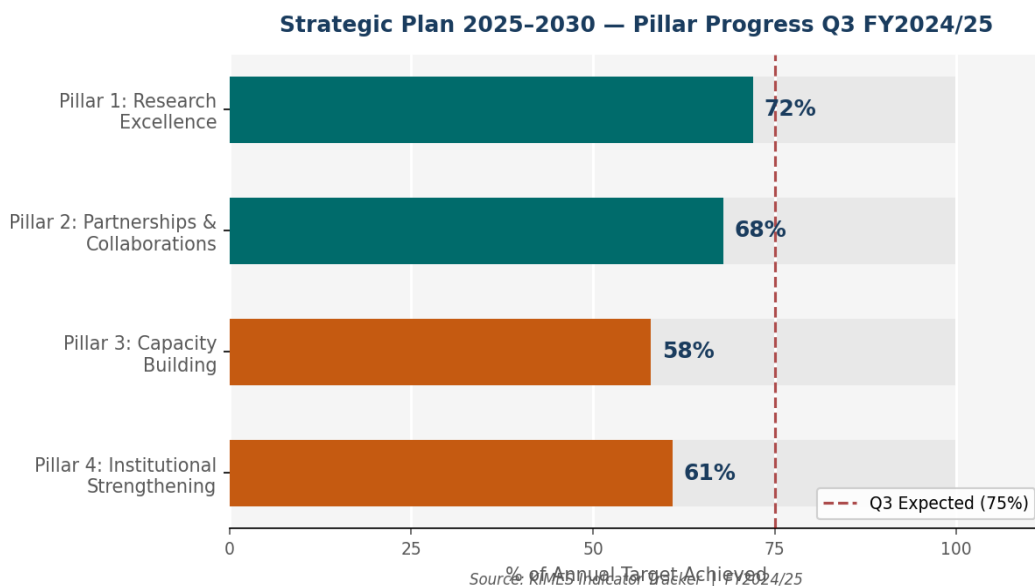


Figure 9.2 — Strategic Plan 2025–2030 Pillar Progress vs Q3 Target (Source: KIMES Indicator Tracker — All Centres)

9.3 Research Output Trends

This multi-line trend chart tracks publications, conference presentations, and policy briefs over 11 quarters. It is used in the Board Annual Report, funder progress reports, and the institutional research profile. It demonstrates trajectory and momentum rather than just a snapshot figure.

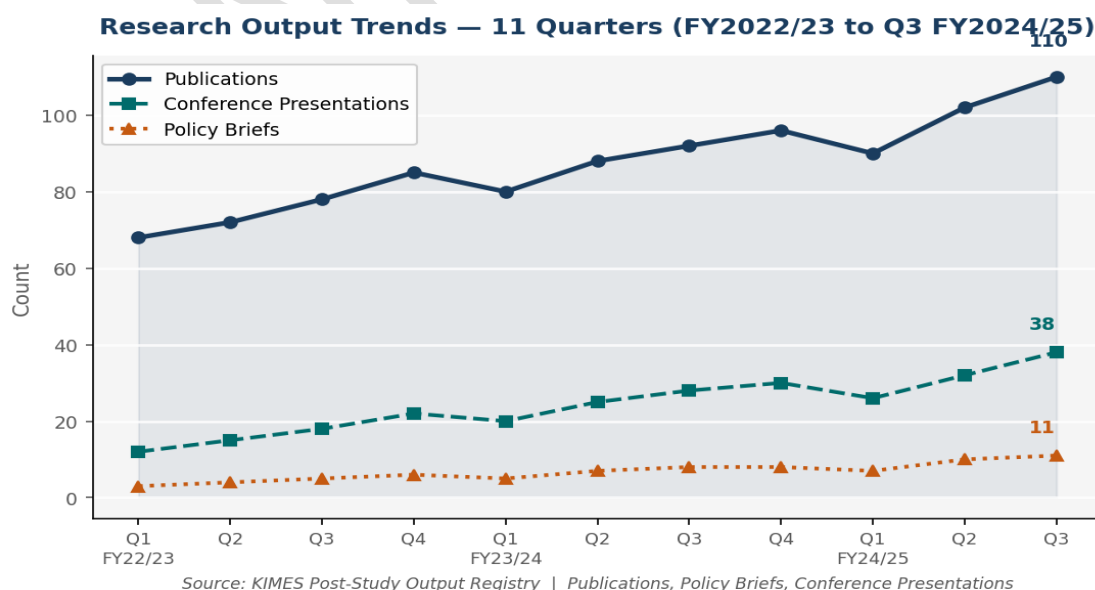


Figure 9.3 — Research Output Trends: Publications, Conference Presentations & Policy Briefs (FY2022/23 to Q3 FY2024/25)



9.4 Financial Performance by Centre

The grouped bar chart compares budget allocations against actual expenditure for each research centre, with the utilisation rate shown as a percentage above each actual bar. Finance Officers and Centre Directors use this chart to monitor burn rates and identify potential under- or over-expenditure risks.

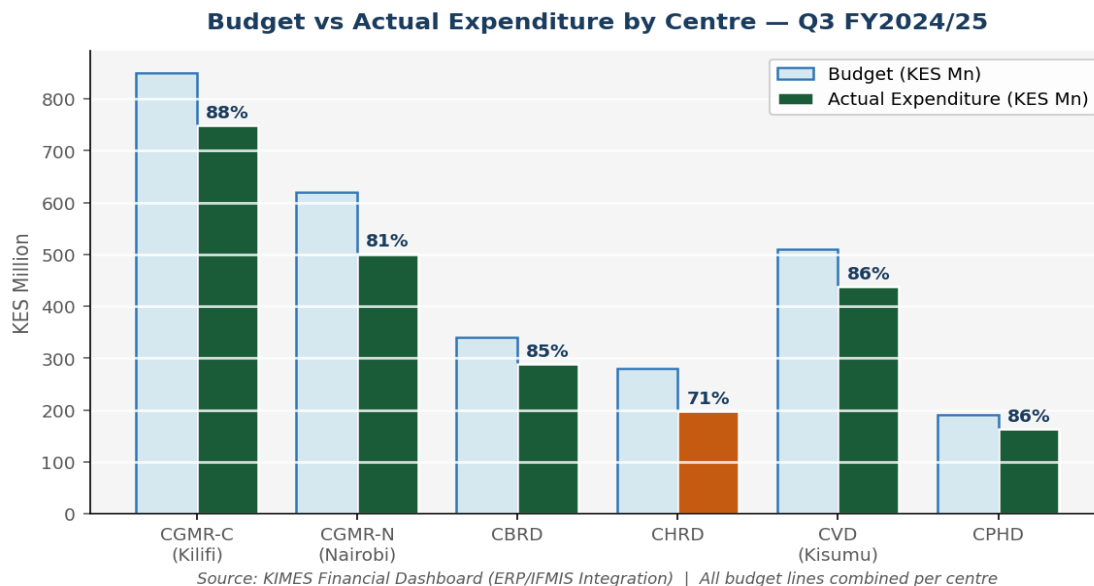


Figure 9.4 — Budget vs Actual Expenditure by Research Centre: Q3 FY2024/25 (Source: KIMES Financial Dashboard — ERP/IFMIS Integration)

9.5 Reporting Compliance Rate by Centre

This chart tracks reporting compliance improvement from baseline (Q1) to current quarter (Q3) by centre, plotted against the 80% institutional target. It is a key institutional performance indicator, reported to the Board quarterly, and serves as a direct measure of KIMES impact since deployment.

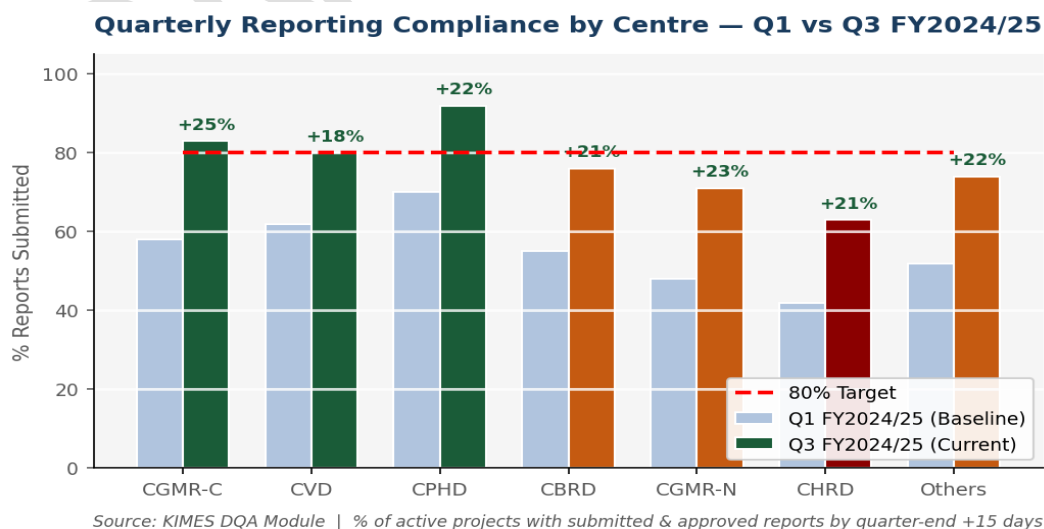
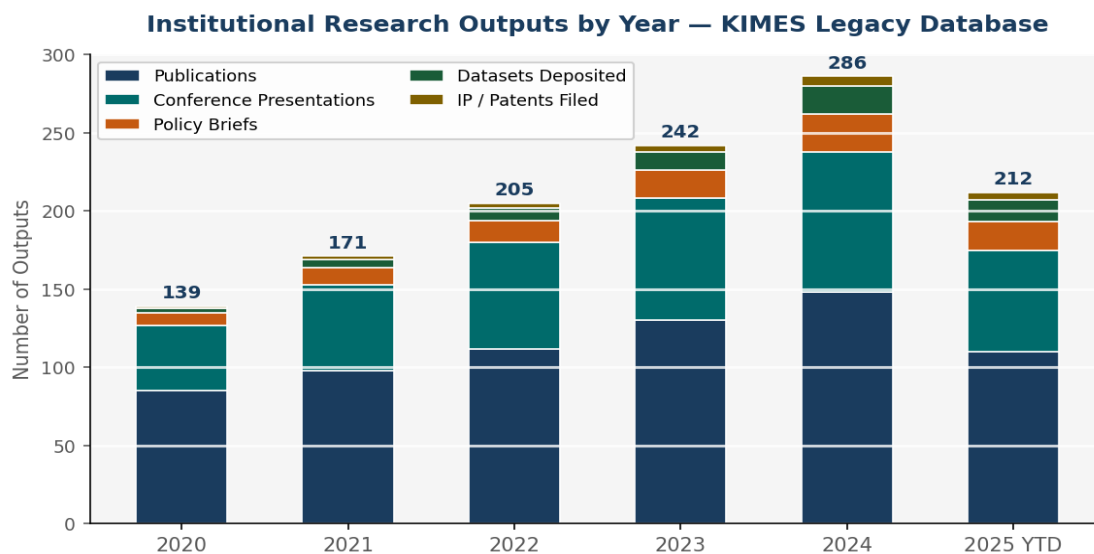


Figure 9.5 — Quarterly Reporting Compliance by Centre: Q1 vs Q3 FY2024/25 Comparison (Source: KIMES DQA Module)



9.6 Post-Study Outputs Registry — Annual Stacked Bar

This stacked bar chart shows the cumulative annual research output tally from the KIMES Post-Study Output Registry, broken down by output type: publications, conference presentations, policy briefs, datasets, and IP/patents. This is the primary visual for demonstrating KEMRI's institutional research contribution to funders, MoH, and the public.



Source: KIMES Post-Study Output Registry | Cumulative annual totals across all centres

Figure 9.6 — Institutional Research Outputs by Year — KIMES Legacy Database (Source: KIMES Post-Study Output Registry — All Centres)



9.7 Procurement Pipeline Status

This bar chart shows the number of active procurement items at each of the seven KIMES-ERP procurement nodes at any given time. It gives the Head of Procurement and Centre Directors instant visibility of where items are in the pipeline and where bottlenecks are forming.

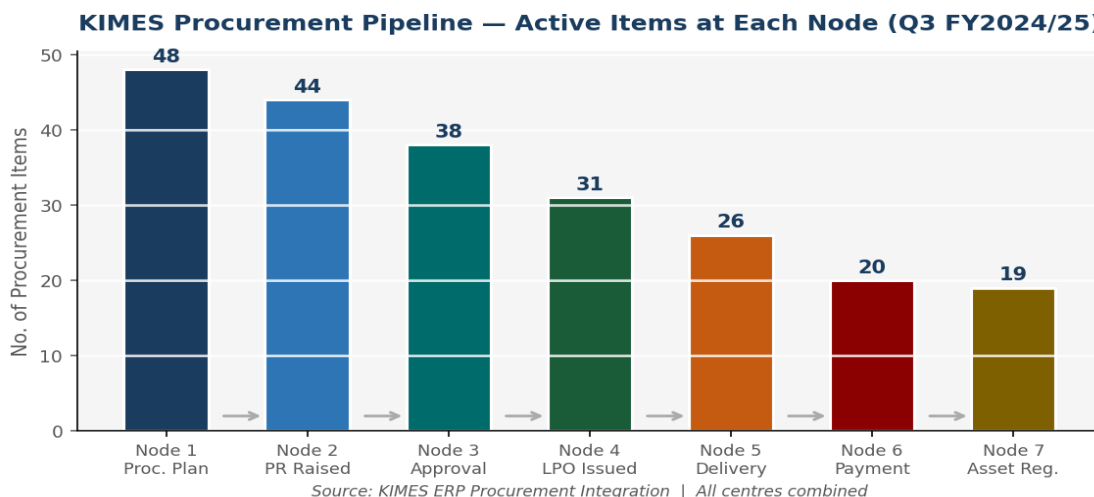


Figure 9.7 — KIMES Procurement Pipeline: Active Items at Each Node (Source: KIMES ERP Procurement Integration — All Centres)

9.8 HR Staffing Plan Compliance — Radar

The radar chart shows HR performance across six dimensions: roles filled vs plan, contract compliance, gender balance, capacity building, staff retention, and training records. This chart is used by the HR Division and the MEL Division Head to identify areas of HR management risk per centre or institution-wide.

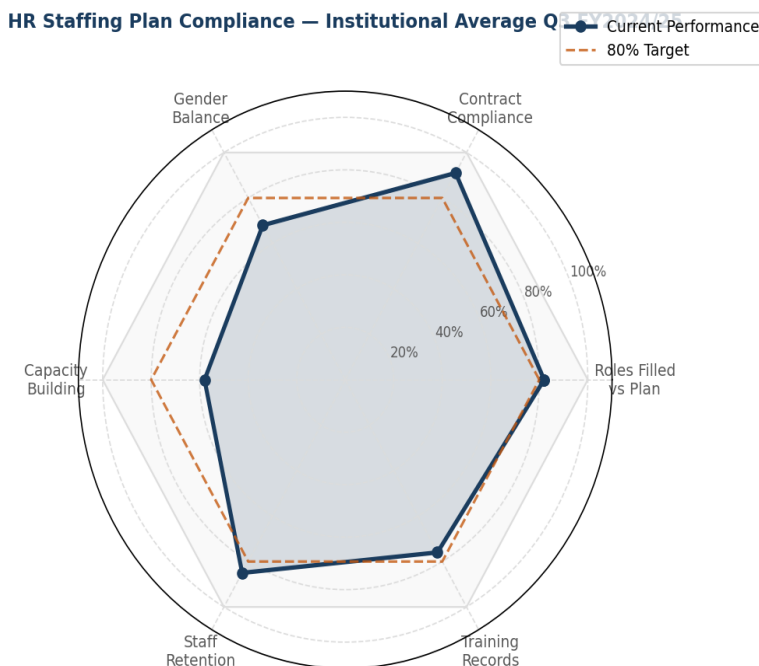


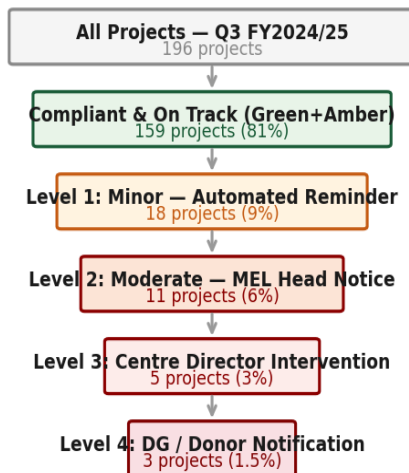
Figure 9.8 — HR Staffing Plan Compliance Radar: Institutional Average Q3 FY2024/25 (Source: KIMES ERP HR Integration)



9.9 Non-Conformity Escalation Funnel

The escalation funnel visualises the distribution of all 196 active projects across the five non-conformity levels — from Compliant & On Track at the top, through the four escalation levels to DG / Donor Notification at the bottom. This visual appears on the Board Scorecard under Compliance & Governance and drives Board attention to the projects at the bottom of the funnel.

KIMES Non-Conformity Escalation Funnel — Q3 FY2024/25



Source: KIMES Escalation Log | Escalation applies progressive institutional authority (MEL Head → Centre Director → DG → Board)

Figure 9.9 — KIMES Non-Conformity Escalation Funnel: Q3 FY2024/25 (Source: KIMES Escalation Log — All Centres)



9.10 AI Reporting Engine — Time Savings

This comparison bar chart quantifies the time savings achieved by the KIMES AI Reporting Engine across all 10 report types, comparing manual production time against AI-assisted production time. The chart demonstrates the financial and operational case for the AI reporting investment and is used in the institutional value-for-money report to funders and the Board.

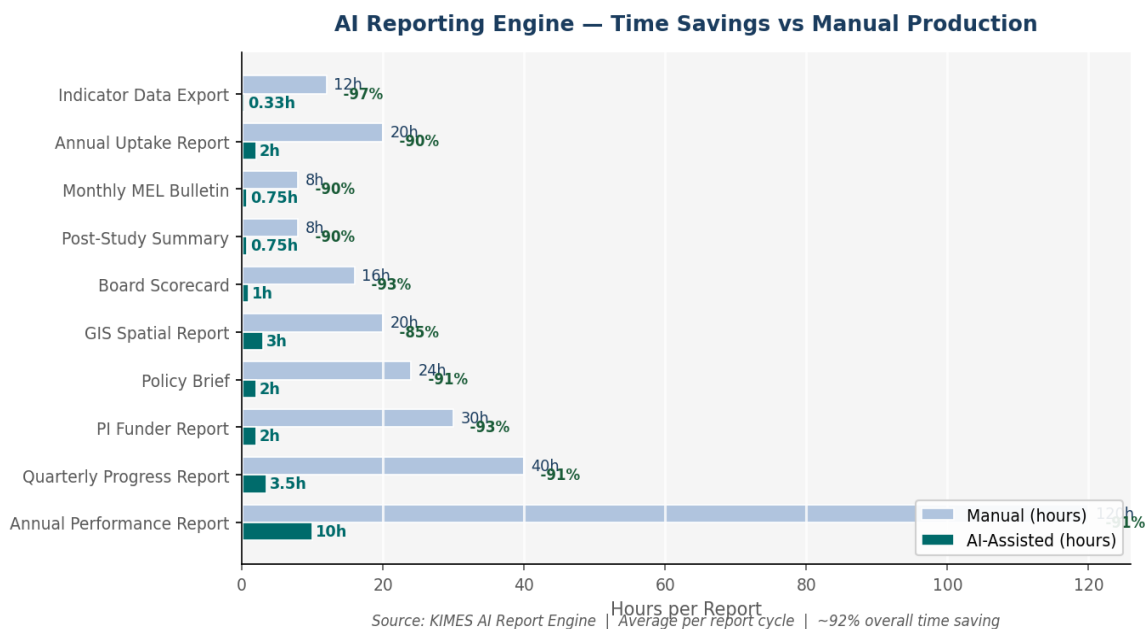


Figure 9.10 — AI Reporting Engine: Manual vs AI-Assisted Production Time per Report Type (~92% Average Time Saving; Source: KIMES AI Report Engine)



10. MANAGEMENT & BOARD DASHBOARD FRAMEWORK

10.1 Dashboard Access Hierarchy

Role	Dashboard View	Key Data Visible	Update Frequency
Principal Investigator	My Projects	Own projects: KPI progress, budget, SERU status, output count, procurement items, staff status, upcoming deadlines, query notifications	Real-time on submission; KPI charts updated quarterly
Centre Director	Centre Portfolio	All projects in centre: RAG dashboard, PI report queue for review, escalation list, centre-level KPI aggregation, output tally, procurement and HR alerts	Real-time; daily digest email
MEL Division Head	Institutional MEL Operations	All centres: DQA scores, reporting compliance rates, escalation log, AI report queue, output registry status, ERP sync status	Real-time dashboard; weekly MEL operations report
Head of Procurement	Procurement Operations	All project-linked procurement items at each node; overdue PRs, LPOs, GRNs; budget commitment rates	Real-time; daily PR/LPO queue
HR Division	Project HR Operations	All project-linked staff: contract end date calendar, extension requests pending, project-funded headcount, capacity risk flags	Real-time; daily contract expiry queue
Deputy Directors	Thematic / Programme View	Cross-centre view by disease area / programme / strategic pillar: KPI trends, output aggregation, financial summary	Real-time; monthly thematic brief
Director General	Executive Dashboard	Full institutional view: all centres, all KPIs, all escalated cases, financial portfolio, Board scorecard preview, donor relationship status	Real-time; DG daily brief (AI-generated, 1 page)
Board Members	Board Scorecard Portal	Quarterly scorecard: strategic plan progress, RAG portfolio, top achievements, items requiring Board attention, comparative trends, donor relationship summary	Quarterly publication; accessible between Board meetings



10.2 AI Reporting Engine — Report Types & Sign-Off Protocol

Report Type	Audience	Manual Time	AI-Assisted	Saving	Sign-Off Required
Quarterly Progress Report (all centres)	Management / Funders (concurrent)	40 hrs	3.5 hrs	91%	Centre Director
Board Performance Scorecard	Board of Management	16 hrs	1 hr	94%	Director General
Monthly MEL Bulletin	Centre Directors / MEL Teams	8 hrs	45 min	88%	MEL Division Head
PI Annual Project Report (funder format)	Funder / Donor	30 hrs	2 hrs	93%	PI + Centre Director + DG Office
Policy Brief (research-to-policy)	MoH / Parliament / Partners	24 hrs	2 hrs	92%	Deputy Director + Centre Director
Indicator Data Export	MEL Division / Finance	12 hrs	20 min	97%	MEL Division Head
Post-Study Output Summary (per PI)	PI / Centre Director / Board	8 hrs	45 min	91%	Centre Director
Annual Research Uptake Report	Board / MoH / Funders	20 hrs	2 hrs	90%	MEL Division Head + DG
GIS Spatial Analysis Report	Management / MoH / Partners	20 hrs	3 hrs	85%	GIS Specialist + Centre Director
Donor Non-Conformity Letter (DG-NCF-001)	Donor / KEMRI Legal	6 hrs	30 min	92%	DG + Legal Counsel (mandatory)
Annual Performance Report (institutional)	Board / MoH / Public	120 hrs	10 hrs	92%	Director General + Board Secretary
TOTAL (per annual cycle)	All stakeholders	304 hrs	25.8 hrs	~92% avg	Human sign-off mandatory for all



11. ENTERPRISE INTEGRATION MAP & 7-YEAR LEGACY TRACKING

11.1 KIMES Enterprise Integration Map

External System	Integration Type	Data FROM System INTO KIMES	Data FROM KIMES TO System	Business Purpose
KEMRI ERP — Finance (IFMIS)	Bidirectional API (daily sync)	Budget vs actual expenditure per project; Payment confirmations; LPO values and commitments	Procurement requisitions; Budget line commitments on LPO issuance	Project financial dashboards; Concurrent donor financial reporting; Budget utilization monitoring
KEMRI ERP — Procurement	Bidirectional API	PR numbers and status; LPO issuance and supplier details; GRN confirmations; Payment release confirmation	Structured procurement requisitions from PI Procurement Plan; Project number and budget line for each PR	Procurement status visibility per project; Equipment registry auto-population; Nodes 1–7 pipeline
KEMRI ERP — HR	Bidirectional API (daily sync)	Staff contract data; Leave records; Payroll charges to project codes; Training records; Off-boarding notifications	Project staffing requests; Contract extension requests (with Centre Director endorsement)	Staffing plan compliance; Contract expiry alerts; Capacity building tally; Gender disaggregation
DHIS2 / KHIS	Read API (quarterly)	County-level disease burden data; Health facility GIS data; National health indicators	KEMRI research coverage data (aggregated, non-identifiable) contributed to national repository	Disease burden maps; Research coverage equity analysis; Policy brief contextualization
REDCap (Clinical Data)	Bidirectional API	Finalized clinical dataset metadata; Protocol amendments; Adverse event reports	Project ID, SERU approval number, and study metadata for REDCap project setup	Linking clinical data to KIMES project record; Output tracking for clinical studies
ODK Central / KoboToolbox	Bidirectional API (real-time)	GPS coordinates from field submissions; Form submission counts and dates; Data collection progress	Study-specific form designs from KIMES project record	GIS layer updates; Field data collection progress monitoring; GPS validation
PubMed / Crossref API	Outbound read API (auto on DOI entry)	Full citation metadata on DOI entry; Quarterly citation count updates	None — read-only integration	Publication registry auto-population; Citation tracking;



External System	Integration Type	Data FROM System INTO KIMES	Data FROM KIMES TO System	Business Purpose
				Impact factor reporting
Scopus / Web of Science API	Outbound read API (quarterly auto-run)	Citation counts; h-index by researcher; Journal quartile rankings	None — read-only integration	Researcher profile metrics; Institutional publication impact reporting
Donor Portal (KIMES-hosted)	KIMES outbound — donor read access	Donor notification preferences	Project dashboard data (read-only); Approved AI-generated reports; Financial summaries; Escalation notices (Level 3+)	Concurrent reporting; Donor transparency; Non-conformity notification
SERU (Ethics Board System)	Bidirectional (semi-automated)	Approval numbers; Expiry dates; Protocol amendment approvals; Adverse event acknowledgements	Protocol amendments requiring SERU review	Ethics compliance monitoring; SERU expiry alerts; Adverse event tracking

11.2 7-Year Output Tracking Timeline per Project

Timeline	Tracking Activity	Auto / Manual	KIMES Action
Project active (Year 0 to N)	Quarterly KPI, financial, staff, lab, equipment, procurement, and HR reports	Manual (PI); auto-validated	Active project record in KIMES
Project closure (Year N)	Study closure report; preliminary output list; final financial reconciliation	Manual (PI)	Transition to Post-Study Tracking status
Year N+1	Publication submission tracking; first datasets deposited; conference presentations	Manual (PI) + DOI auto-import	Post-Study Output Registry created
Year N+2	Publication citation count tracking begins (auto-quarterly); policy brief logging; IP stage update	Auto (Crossref/Scopus) + Manual (PI)	Citation tracker activated per publication DOI
Year N+3	Mid-term output audit by MEL Division: completeness check of output registry	Auto-prompt to PI + MEL Officer audit	Completeness score calculated; missing items flagged
Year N+4 to N+7	Annual AI policy citation scan; citation counts auto-updated; IP expiry monitoring; Research Uptake Score updated	Auto (AI scan + DOI resolution)	Annual Research Uptake Score updated; IP alerts sent



Timeline	Tracking Activity	Auto / Manual	KIMES Action
Year N+7	Full research lifecycle report auto-generated: all outputs, citations, policy uptake, IP, capacity outputs linked to project	Auto (AI-generated)	Final institutional research contribution record archived

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12. CONSOLIDATED WORKFLOW REFERENCE TABLE

This one-page summary provides a quick reference for all 15 workflow steps plus the non-conformity escalation levels, showing phase, responsible party, M&E Officer role, and key output at a glance.

Step	Phase	Title	Primary Responsible	M&E Officer Role	Key Output
1	Pre-Study	Project registration	PI + Centre Director (approver)	Admin support, project ID generation	KIMES Project ID + Logframe + Donor Portal record
2	Pre-Study	KPI & milestone definition	PI + Centre Director (approver)	Technical support (SMART formatting)	Approved monitoring plan + Donor reporting calendar
3	Active	Quarterly milestone reporting	PI (solely responsible)	Reminders, pre-fill support	Quarterly report + auto-CC to KEMRI + Donor Portal queue
4	Active	Automated DQA	KIMES System (automated)	Monitor DQA; follow up on flagged fields	DQA score + field-level flags
5	Active	Scientific peer review & RAG	Centre Director / Senior Scientist	Prepare data summary for Centre Director	Approved report + RAG status + Donor Portal update
6	Active	Escalation (non-reporting)	MEL Head → Centre Director → DG → Board	Initiate escalation log	Formal notices + Board flag (see Section 7 for donor escalation)
7	Active	AI report generation	KIMES AI + Centre Director (sign-off)	Fact-check AI draft; route for sign-off	Word + Excel + PDF + Donor Portal concurrent report
8	Active	Donor concurrent report routing	PI (primary) + Grants Office / MEL Division (routing)	Donor Portal management; report calendar	Internal CC + Donor Portal update + compliance record
9	Active	GIS & spatial update	GIS Specialist	Data sync support	Updated maps + spatial reports
10	Post-Study	Study closure report	PI + Centre Director (approver)	Transition to post-study tracking	Approved closure report + Post-Study Registry created



Step	Phase	Title	Primary Responsible	M&E Officer Role	Key Output
11	Post-Study	Abstract & conference capture	PI (self-report)	Output registry management	Conference output record
12	Post-Study	Publication & citation tracking	PI (on submission) + Auto (on DOI)	Registry management; DOI import monitoring	Publication record + quarterly citation auto-update
13	Post-Study	Dataset & tools registry	PI + Data Mgmt Officer	FAIR compliance verification	Dataset deposit record + FAIR score
14	Post-Study	Policy brief & uptake tracking	PI + MEL Division (annual AI scan)	Annual AI citation scan; uptake log	Policy engagement record + Annual Uptake Score
15	Post-Study	IP & patents registry	PI + Innovation Unit + Legal/IP Office	IP registry maintenance; expiry alerts	IP portfolio record + annual IP report
L1–L4	Escalation	Non-conformity escalation	KIMES (L1) → MEL Head (L2) → Centre Director (L3) → DG (L4)	Log management; notice routing; audit trail	Graduated escalation notices + Donor notification letter DG-NCF-001 (L4 only)



13. IMPLEMENTATION ROADMAP

Phase	Period	Key Deliverables	Success Criteria
Phase 0 PoC Validation	May–Jul 2025	Present this document to KEMRI MEL Division, Centre Directors, and a representative sample of PIs (minimum 5 from different centres). Gather structured feedback on: workflow logic, governance structure, review authority, donor liaison protocol, ERP integration feasibility, and non-conformity escalation acceptability. Revise based on feedback. Board approval to proceed.	Board resolution to proceed; PI acceptance of peer-review governance model; Centre Director and Finance/HR/Procurement buy-in on ERP integration architecture
Phase 1 Foundation	Aug–Nov 2025	Procure and configure server infrastructure. Build and deploy core KIMES modules: Project Registry, KPI Tracker, Output Registry, Document Store, DQA Engine, role-based dashboards. Configure KIMES Donor Portal. Begin data migration for all active projects. Staff training (10-module curriculum). Grant onboarding protocol and Grant Agreement Data Addendum finalized by Legal.	KIMES operational with all active projects registered; 200+ user accounts; automated DQA live; Centre Directors and PIs trained; Legal instruments approved
Phase 2 Workflow Go-Live	Dec 2025–Mar 2026	Full workflow activated for all active projects. First quarterly reporting cycle through KIMES. Donor Portals configured for top 20 funders by grant value. Post-Study Output Registry opened for all projects closed in last 3 years (retrospective entry). GIS portal live. First AI-assisted quarterly report and first concurrent donor report generated and approved.	100% of active projects in KIMES; first quarterly cycle >80% compliance; first AI report signed off; first concurrent donor reports delivered; Donor Portal live for top 20 funders
Phase 3 ERP & AI Integration	Apr–Sep 2026	KIMES-ERP Procurement and HR bidirectional APIs go live (Nodes 1–7 and HR-1 to HR-7). AI Reporting Engine fully live for all 11 report types. DHIS2, PubMed/Crossref, and Scopus API integrations live. First Board Quarterly Scorecard auto-generated. Annual policy citation AI scan first run. Non-conformity escalation protocol officially in effect.	All ERP integrations live and validated; all AI report types operational; first Board Scorecard delivered; escalation protocol tested and approved by DG and Legal
Phase 4 Optimization & Legacy	Oct 2026–2030	Continuous improvement based on user feedback. Retrospective output data entry for KEMRI publications 2015–2024. Annual system performance audits. Mid-term strategic plan review (2027). AI model fine-tuning for KEMRI institutional voice. Full 7-year legacy database building. Annual Research Uptake Scores published each October.	≥90% user satisfaction; output registry covers 2015–2030; Research Uptake Scores published annually; KEMRI recognized as continental MEL and data governance leader



14. SIGNATORIES & DOCUMENT CONTROL

Prepared by:

Pathway Technologies

Lead Systems Architect / PoC Lead | May 2026

Signature: _____

Date: _____

Endorsed by:

Director General, KEMRI

Kenya Medical Research Institute | May 2026

Signature: _____

Date: _____



*KIMES v5.0 — Complete System Design with Dashboards & Visualizations | Pathway Technologies for KEMRI | May 2026
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